VIVA PREPARATION GUIDE

Agentic AI for Automated Digital Marketing
Campaign Generation, Scheduling, and Management

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How to use this document: Read each question aloud, then check your answer against the key points. If you can cover all bullets confidently in under 2 minutes, you are ready. Flag the questions you struggle with and revisit them the day before.

SECTION 1 — Personal & Reflective (These came up in the 2025 batch)

Q1. What is the best part of your project?

Key points: Pick one thing and own it — do not list everything.

- The dual-orchestration architecture: implementing both sequential and hierarchical patterns in the same codebase under the same LLM models so that any quality difference is genuinely attributable to the pattern, not to different tools
- End-to-end automation — brand description to a ready-to-review 7-day Instagram calendar with AI images, in under 90 seconds
- The evaluation framework: using LLM-as-a-judge (Zheng et al., 2023) with a structured five-dimension rubric designed specifically for creative content — no benchmark for this existed before

💬 ***Suggested opening:*** *"The strongest part for me is the experimental design. By running both orchestration patterns on the same brand prompts, with the same models and the same evaluation rubric, I can attribute quality differences directly to the architecture rather than any external variable. That level of control doesn't exist in prior work on multi-agent marketing systems."*

Q2. What was the biggest obstacle you faced and how did you resolve it?

Key points: Be specific — one real technical problem is more convincing than a vague answer.

- Problem: The LangGraph graph (graph.py) was built correctly but the API routes were calling the ContentAgent class directly, bypassing the graph entirely — two parallel code paths doing the same job
- How found: Noticed during a code review that campaign_graph was never invoked in routes.py
- Resolution: Refactored routes.py to call the compiled graph as the single entry point, removing the ContentAgent shortcut — this also made adding the hierarchical mode straightforward since the graph topology is now the single source of truth
- Second obstacle worth mentioning: Port 8000 blocked by a system process on Windows — resolved by switching both backend and frontend to port 8001

💬 ***Suggested opening:*** *"The most significant technical obstacle was discovering that my LangGraph graph was not actually being called in production — the API routes were bypassing it and calling a content agent class directly. I found this during a code review, and resolving it required refactoring the routes to treat the compiled graph as the canonical entry point. It was a useful lesson in not assuming that code that exists is code that runs."*

Q3. What would you do differently if you started again?

Key points: Show self-awareness — do not say "nothing", do not be too self-critical.

- Wire the LangGraph graph into the API from day one — the shortcut of using ContentAgent directly created technical debt that needed undoing later
- Design the InteractionLog database table before building the HITL interface, not after — adding logging retroactively required touching every endpoint
- Apply for Meta Graph API access on week one — the approval process takes time and it blocked the scheduling module

💬 ***Suggested opening:*** *"I would implement the database schema for interaction logging before writing the HITL interface, rather than adding it afterwards. Designing the data model first would have* *made every approve, reject and regenerate endpoint cleaner because logging would have been a first-class concern from the start rather than a retrofit."*

Q4. What will you do to improve this project in the future?

Key points: This is in your report — reference it. Three concrete items max.

- Reinforcement learning from human feedback: the interaction logs capture before/after content pairs that form a preference dataset — this could be used to fine-tune the caption model and reduce intervention rates over time
- Multi-platform extension: the platform field in the data model already supports it; adding LinkedIn, Facebook, and X requires only platform-specific Content Agent prompt variants and their respective API clients
- WhatsApp conversational editing: users receive campaign drafts as messages and approve or request changes via natural language reply — makes the system accessible to SMB users unlikely to log into a web dashboard

💬 ***Suggested opening:*** *"The most impactful next step would be closing the feedback loop — using the interaction logs as a preference dataset to fine-tune the caption generation model. Right now human corrections improve one campaign; with RLHF they would improve every future campaign."*

Q5. Why did you choose this topic?

Key points: Connect the personal motivation to the research gap — do not just say "I like AI".

- The gap was clear and specific: no existing tool combines strategy, content generation, image creation, HITL review, and scheduling in a single agentic workflow
- The research questions were unanswered: sequential vs. hierarchical orchestration had never been empirically compared in a creative domain
- Practical impact: SMBs spend disproportionate resources on social media content — a working system addresses a real business problem

💬 ***Suggested opening:*** *"I was drawn to the specific gap — not just that AI could help with marketing, but that nobody had empirically compared orchestration patterns in a creative domain. All existing benchmarks use coding tasks where correctness is binary. Marketing quality is subjective and multidimensional, and that made the comparison genuinely novel."*

Q6. How confident are you in your results?

Key points: Acknowledge limitations honestly — examiners respect self-awareness more than overconfidence.

- Confident in the system functionality — the pipeline is working end-to-end
- Acknowledge LLM-as-a-judge limitation: the evaluator may assess brand consistency differently from a practising marketer
- Reliability check between LLM judge and researcher scores partially validates the approach
- Small sample size (2–3 human evaluators) limits statistical power — acknowledge this directly

💬 ***Suggested opening:*** *"I am confident in the directional findings, but I am careful about the magnitude. The LLM-as-a-judge approach is methodologically grounded in Zheng et al. but has not been validated specifically for marketing assessment. I treat the results as indicative rather than conclusive, and I flag replication with professional evaluators as the first priority for future work."*

SECTION 2 — Research Questions & Literature

Q7. Explain your two research questions and why they matter.

- RQ1: Does sequential or hierarchical orchestration produce better quality marketing campaigns? — matters because no empirical comparison exists in creative domains; all prior benchmarks use coding tasks
- RQ2: Does human-in-the-loop feedback measurably improve AI-generated content? — matters because it quantifies the value of human oversight, which is assumed but not measured in existing systems
- Together they provide both architectural guidance (which pattern to build) and operational guidance (how much human review is worth)

Q8. What is the difference between sequential and hierarchical orchestration? What are their failure modes?

- Sequential: fixed linear chain — Strategy Node output feeds Content Node. Predictable, auditable, easy to debug
- Failure mode — semantic drift: a misinterpretation by the Strategy Node propagates uncorrected through every downstream node
- Hierarchical: an Orchestrator Agent decomposes the brief into parallel sub-tasks (brand strategy, posting schedule, visual guidelines), specialist agents execute them, orchestrator synthesises, then Content Node generates
- Failure mode — orchestrator overhead: every delegation and synthesis step adds latency; orchestrator is a single point of failure
- LangGraph shared state object mitigates overhead by letting all agents read from a common state rather than passing data through the orchestrator

Q9. Why did you choose LangGraph over AutoGen, CrewAI, or MetaGPT?

- LangGraph: supports both sequential (DAG) and cyclic graph topologies in the same codebase — the only framework that lets you switch between sequential and hierarchical by changing the graph definition
- Explicit state management via CampaignState: all agents read and write a single typed object, preventing the data inconsistency problems that arise in AutoGen's conversation-thread model
- AutoGen: flexible but unstructured outputs — difficult to guarantee parseable JSON for the strategy plan
- MetaGPT: role assignments designed for software engineering, not marketing — constrains creative flexibility
- CrewAI: developer-friendly but limited peer-reviewed validation; no cyclic execution support

Q10. What is the ReAct framework and how does it relate to your work?

- ReAct (Yao et al., 2023): interleaves Reasoning (internal thought) with Acting (external tool call) in an iterative loop
- Establishes the architectural pattern that modern frameworks adapt — thought → action → observation → thought

- Your sequential pipeline follows the ReAct-inspired pattern: Strategy Node reasons about the brand brief and acts by producing a plan; Content Node reasons about the plan and acts by producing captions and image prompts
- Key difference: your system uses a multi-node graph rather than a single looping agent

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Q11. Why are BLEU and ROUGE inappropriate for evaluating your system's output?

- BLEU and ROUGE measure lexical overlap with a reference text — they reward outputs that use the same words as a human-written example
- Marketing captions have no single correct reference: two equally good captions for the same brand may share almost no words
- Brand alignment, strategic coherence, and visual relevance are semantic qualities invisible to lexical metrics
- LLM-as-a-judge (Zheng et al., 2023) and expert rubric assessment evaluate meaning and quality rather than word overlap

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Q12. What is Cohen's kappa and why did you use it?

- Cohen's kappa measures inter-rater agreement adjusted for chance — unlike percentage agreement, it corrects for the possibility that raters agree randomly
- Used to validate that your evaluators (human or LLM) are applying the rubric consistently rather than scoring arbitrarily
- Kappa > 0.6 is generally considered substantial agreement; > 0.8 is near-perfect
- Important because your evaluation dimensions (brand consistency, strategic coherence) are subjective — kappa quantifies whether the rubric is making the assessment reproducible

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Q13. What did Amershi et al. (2019) contribute and how does it appear in your interface design?

- Amershi et al. identified 18 design principles for human-AI collaboration — most influential: make uncertainty visible, support efficient correction, and scope AI actions conservatively
- Make uncertainty visible: each post card labels content as AI-generated
- Support efficient correction: all five HITL actions (approve, reject, edit caption, regenerate caption, regenerate image) are accessible directly from the post card — no navigation required
- Scope conservatively: all posts default to "pending" — nothing is published without explicit approval; no bulk-approve function

SECTION 3 — Architecture & Design

Q14. Walk me through what happens when a user submits a brand brief.

- 1. User fills the Brand Input Form in React and clicks Generate
- 2. POST /api/calendars — FastAPI creates a Calendar record (status: generating) and returns 201 immediately
- 3. Background task starts: LangGraph graph invoked with CampaignState populated from brand input
- 4. Strategy Node calls Claude → produces 7-day campaign plan (theme, message, CTA, posting time per day)
- 5. Content Node calls Claude → produces caption + image prompt per day
- 6. Image generation: DALL-E 3 called for each post in parallel (max 3 concurrent), URLs saved to Post records
- 7. Calendar status updated to "ready"
- 8. Frontend polls GET /api/calendars/:id every 3 seconds, detects status change, renders review interface

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Q15. Why FastAPI over Flask or Django?

- Async-native: background tasks and concurrent image generation require async/await — FastAPI supports this natively; Flask does not without extensions
- Automatic OpenAPI schema generation from Pydantic models — no manual documentation needed
- Pydantic v2 integration: runtime validation of LLM JSON outputs before they reach the database
- Django is too heavyweight for a single-domain API — its ORM, admin, and template engine add complexity without benefit here

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Q16. What is CampaignState and why is it important?

- CampaignState is a TypedDict that defines all fields that flow between LangGraph nodes: brand inputs, generated campaign plan, list of post specifications
- Every node receives the full state and returns an updated version — no node communicates directly with another
- This design means adding a new node (e.g., a Scheduler Node) requires only defining what it reads from state and what it writes back — no changes to other nodes
- In the hierarchical mode, the shared state lets three parallel sub-agents each write their outputs to different state fields, which the Orchestrator then reads during synthesis — no orchestrator bottleneck for data passing

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Q17. What happens if the Claude API call fails mid-generation?

- The background task wraps the entire generation in a try/except block
- On any exception, the Calendar status is set to "error" and the exception is logged
- The frontend detects the "error" status on its next poll and displays an error state to the user
- Current limitation: there is no automatic retry logic — a failed generation requires the user to delete and recreate the calendar
- Future improvement: exponential backoff retry on API rate-limit errors (HTTP 429)

⚠ **Avoid:** Do not claim the system is fault-tolerant — it handles failures gracefully but does not self-heal.

Q18. Why SQLite for the database? Would that work in production?

- SQLite: zero-config, file-based, appropriate for development and single-user local testing
- SQLAlchemy ORM abstracts the database entirely — switching to PostgreSQL requires only changing the connection string and running a schema migration
- Not suitable for production with concurrent users: SQLite uses file-level locking, so concurrent writes from multiple users would cause contention
- PostgreSQL would be the production choice: row-level locking, connection pooling, better performance under load

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Q19. How does your polling mechanism work and what are its drawbacks?

- Frontend useEffect hook calls GET /api/calendars/:id every 3 seconds while status === "generating"
- Interval is cleared when status transitions to "ready" or "error"
- Drawback 1: generates unnecessary HTTP requests — 30 requests for a 90-second generation
- Drawback 2: 3-second lag between completion and UI update
- Better approach: WebSockets or Server-Sent Events — server pushes a single notification when generation completes. Polling was chosen for simplicity within the project timeline

SECTION 4 — AI Models & Prompt Engineering

Q20. Why Claude for captions and DALL-E 3 for images? Why not use one provider for both?

- Claude: strong instruction-following and brand tone adherence; low hallucination rate on constrained prompts; reproducible with temperature=0
- DALL-E 3: superior prompt adherence compared to Stable Diffusion — fewer irrelevant outputs for structured marketing prompts; accurate text rendering within images
- Could use GPT-4o for text and DALL-E 3 for images (same provider) — the LLM factory pattern in llm.py makes this a one-line change
- Stable Diffusion alternative: free to run locally, but brand consistency across multiple images is harder to control without fine-tuning or ControlNet

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Q21. What is prompt sensitivity and how did you address it?

- Prompt sensitivity: small changes in prompt wording produce substantially different outputs — a risk when brand consistency across 7 days is required
- Mitigation 1: few-shot example of target caption style included in the Content Agent prompt
- Mitigation 2: brand tone and name repeated in both system and user messages
- Mitigation 3: structured JSON output format requested with explicit field definitions — reduces the model's degrees of freedom
- Mitigation 4: the Strategy Node produces a campaign plan first, giving the Content Node richer context than a raw brand description alone

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Q22. How do you make your image prompts produce brand-consistent results?

- Image prompts are generated by the Content Node — not typed by the user — and encode: brand name, product category, colour palette instruction, composition style (e.g., flat lay, lifestyle shot), and negative prompt
- Negative prompt excludes: watermarks, text overlays, people (unless brand brief specifies), blurry backgrounds
- DALL-E 3 is called with quality=hd for all images — ensures consistent resolution across the campaign
- Limitation: text prompts cannot guarantee adherence to a specific hex colour palette or product geometry — motivates future use of reference image conditioning

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Q23. What is RLHF and how does it relate to your work?

- Reinforcement Learning from Human Feedback (Ouyang et al., 2022): models are fine-tuned using human preference data — a human labels which of two outputs is better, and the model is updated to prefer the winner
- Your interaction logs capture exactly this: before/after content pairs for every edit and regeneration action, implicitly labelling the after version as preferred
- You do not implement RLHF in this project — but the logging infrastructure collects the preference dataset that would make it possible
- Identified as the highest-impact future work item

SECTION 5 — Evaluation & Results

Q24. What is LLM-as-a-judge and why is it valid here?

- Zheng et al. (2023): demonstrated that GPT-4 as an evaluator achieves high agreement with human preferences on open-ended quality tasks — validated on MT-Bench and Chatbot Arena
- Used in place of fully recruited expert evaluators to assess 20 campaigns against the five-dimension rubric
- Validity argument: the rubric makes the scoring criteria explicit and consistent — the LLM judge applies the same criteria every time, eliminating human fatigue and inconsistency
- Limitation: the LLM judge may weight brand consistency differently from a practising marketer — reliability check with researcher and peer scores partially addresses this

⚠ **Avoid:** Do not claim LLM-as-a-judge is equivalent to human expert evaluation — say it is a validated methodology with acknowledged domain-specific limitations.

Q25. How did you evaluate RQ1 — the orchestration pattern comparison?

- Same 10 brand prompts processed through both configurations — 20 campaigns total
- Brand prompts span 5 industries (furniture, fashion, food, fitness, technology) to test generalisability
- LLM judge scores each campaign blind on 5 dimensions: brand consistency, content quality, strategic coherence, visual relevance, overall effectiveness (1–5 Likert)
- Automated metrics: content diversity score, Flesch-Kincaid readability, hashtag semantic similarity, generation latency
- Statistical test: paired t-test or Wilcoxon signed-rank on expert scores between configurations

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Q26. How did you evaluate RQ2 — the human-in-the-loop impact?

- 2–3 participants each generated one campaign and completed a full review session using the HITL interface
- Interaction logs recorded every approve, reject, edit caption, regenerate caption, and regenerate image action with timestamps and before/after content
- LLM judge scored the AI-only output (no human review) and the post-HITL final campaign for the same brand — quality delta measures HITL impact
- Baseline comparison: AI-only campaign vs. post-HITL campaign vs. manually created campaign using a template
- Participants completed SUS questionnaire — System Usability Scale, 0–100

Q27. What does the SUS score tell you?

- System Usability Scale (Brooke, 1996): 10 statements rated 1–5, converted to 0–100 score
- Score above 68 indicates above-average usability; above 80 is excellent
- Provides a standardised, comparable usability benchmark that can be cited in future work
- [INSERT your actual score here before the viva]

SECTION 6 — Legal, Social, Ethical & Professional Issues

Q28. How does your system comply with GDPR?

- No personal consumer data is collected at any point — the system processes brand descriptions and AI-generated content only
- API keys stored in environment variables, never committed to version control
- Evaluation participant data (interaction logs, SUS responses) anonymised and stored locally — not shared with third parties
- Meta Graph API integration uses OAuth — no Instagram credentials are stored by the application

Q29. What are the ethical implications of AI-generated marketing content?

- False claims risk: AI may generate product benefits or statistics not supported by the brand — addressed by guardrails in the Strategy Agent prompt
- Disclosure: UK ASA/CAP Code and Instagram policies require AI-generated content is not presented misleadingly — system encourages review and personalisation before publishing
- Intellectual property: legal status of AI-generated images is unsettled in UK law — documentation advises users to seek guidance before commercial use
- Manipulation risk: AI could generate psychologically manipulative copy — Content Agent prompt explicitly prohibits this

Q30. Who owns the content generated by your system?

- Unsettled area of UK law — the Copyright, Designs and Patents Act 1988 Section 9(3) assigns authorship of computer-generated works to "the person who made the arrangements"
- Likely the user (business) owns the content, but this has not been tested in court for LLM outputs
- DALL-E 3 terms of service: OpenAI assigns ownership of generated images to the user, subject to their usage policy
- Claude terms: Anthropic assigns output to the user
- System documentation acknowledges uncertainty and advises legal guidance for commercial use

SECTION 7 — Limitations & Difficult Questions

Q31. Your system only targets Instagram — does that limit the contribution?

- Scope restriction is intentional and justified: the proposal defines a Minimum Viable Research Core that prioritises depth over breadth
- The research questions (orchestration pattern comparison, HITL impact) are platform-agnostic — findings apply to any content type, not only Instagram
- The architecture already supports multiple platforms via the platform field in the data model — extension is a matter of adding platform-specific prompt variants, not redesigning the system
- Prior work on multi-agent marketing systems does not exist for any platform — Instagram as a starting point still represents a novel contribution

⚠ **Avoid:** Do not be defensive — agree that multi-platform is future work, then redirect to the research contribution.

Q32. How do you know your LLM-generated captions are actually good?

- Three independent quality signals: LLM-as-a-judge rubric scores, automated metrics (readability, hashtag relevance), and HITL intervention rate
- High approval rate without editing is indirect evidence of quality — if captions were poor, users would edit or regenerate more frequently
- Pre/post HITL quality delta: if LLM judge scores improve after human review, the before-review score reflects the genuine autonomous quality level
- Acknowledged limitation: no ground truth — there is no objectively correct Instagram caption, only human preferences

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Q33. What if your evaluators were biased because they knew you?

- Acknowledged as a limitation of convenience sampling in Section 6.2 of the report
- Mitigation: blind assessment — evaluators see "Campaign A" and "Campaign B", not which orchestration mode produced each
- Pre-screening questionnaire confirmed minimum social media familiarity before participation
- The LLM-as-a-judge provides a non-human baseline score for comparison
- Replication with professional marketing evaluators is explicitly recommended as future work

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Q34. What is the scalability of your system?

- Current bottleneck: SQLite file-level locking — not suitable for concurrent multi-user production use
- Image generation: DALL-E 3 rate limits mean 7 images per campaign takes 30–60 seconds with 3 concurrent requests
- LLM calls: Claude API rate limits could cause delays for high-volume use
- Migration path: SQLite → PostgreSQL (connection string change), add a task queue (Celery + Redis) for image generation jobs, add caching for repeated brand prompts
- Frontend polling: replace with WebSockets for lower latency and fewer HTTP requests

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Q35. How is your work different from just using ChatGPT to write Instagram posts?

- ChatGPT generates individual posts on demand — it has no concept of a multi-day campaign strategy, no image generation integration, no scheduling, and no structured review workflow
- Your system: Strategy Agent first generates a 7-day coherent campaign plan (not just a post); Content Agent uses that plan to ensure strategic consistency across the week; DALL-E 3 generates matching visuals; HITL interface manages the review workflow; scheduler publishes to Instagram
- Research contribution: comparative evaluation of two orchestration architectures in a creative domain — ChatGPT usage has no architectural evaluation or empirical comparison

⚠ **Avoid:** This question tests whether you understand the novelty of your contribution. Be specific about what a single LLM chat session cannot do.

QUICK REFERENCE — Key Terms & People

Know these cold — examiners often ask "can you explain X in one sentence"

Term / Person	One-line definition
LangGraph	Python framework that models LLM workflows as directed graphs with explicit shared state management at each node
CampaignState	TypedDict object that carries all data between LangGraph nodes — the single source of truth for one generation run
ReAct	Yao et al. (2023) — agent pattern that interleaves Reasoning (thought) with Acting (tool call) in a loop
AutoGen	Wu et al. (2023) — multi-agent framework where agents communicate through structured conversation threads
MetaGPT	Hong et al. (2024) — assigns professional role identities to agents; strong on structured outputs
Semantic drift	Your term — error mode of sequential pipelines where upstream misinterpretation compounds downstream
Orchestrator overhead	Your term — latency and failure-point cost of hierarchical delegation through a coordinator agent
LLM-as-a-judge	Zheng et al. (2023) — using a capable LLM (e.g., GPT-4o) to score outputs against a rubric in place of human evaluators
RLHF	Ouyang et al. (2022) — fine-tuning LLMs using human preference labels; motivates your interaction logging
Amershi et al.	(2019) — 18 principles for human-AI interaction design; underpins your HITL interface decisions
Cohen's kappa	Inter-rater agreement measure adjusted for chance; used to validate evaluator consistency
SUS	System Usability Scale (Brooke, 1996) — 10-item questionnaire producing a 0–100 usability score
Flesch-Kincaid	Automated readability metric — adapted here for caption length to measure content accessibility
Pydantic v2	Python data validation library used to validate LLM JSON outputs before database writes
DALL-E 3	OpenAI image generation model used for creating marketing visuals from structured text prompts

Final tip: if you do not know the answer to a question, say "I would need to think about that more carefully — what I can say is..." and pivot to something you do know. An examiner who sees you think honestly scores higher than one who sees you bluff.